

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



447/1

Power Mechanics

Paper 1

Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

a. Candidates should have the following for their examination.

Drawing Instruments and drawing paper size A4.

b. Section A- Answer ALL questions(40marks)

c. Section B- Answer Question 11 and any other three questions. (60marks)

SECTION A (40MARKS)

1 a)List three major components of the diesel fuel system **(3marks)**

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b) Arrange the following personnel training levels in the correct order starting with the highest,
(2marks)

Technician, Craftsman, Technologist, Artisan

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2 i) Differentiate between live and dead axle. **(2marks)**

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ii) Explain the following terms as used in the suspension system. **(2marks)**

a) sprung weight

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b) un sprung weight

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3 i) State two reasons for alloying metals. **(2marks)**

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ii) For each of the following components state two measurements which should be taken during inspection.
(2marks)

a) Crankshaft

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b) Piston rings

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4 State one use of each of the following tools (2marks)

a) Piston ring compressor

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b) Micrometers

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c) Stethoscope

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d) Compression gauge

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b) Differentiate between friction and antifriction bearings. Give one example of each type.(4mks)

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5 a) State one cause of each the following excessive tire tread wear patterns . **(1mark)**

i) feathered edge

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ii) shoulder wear

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e) State four physical checks that need to be carried on a car battery. **(2marks)**

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6 a) State two possible causes of a clutch slip. **(1mark)**

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b) Give two reasons why bearings are necessary. **(2marks)**

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c) State two requirements for combustion in a diesel engine **(2marks)**

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7 a) State two possible causes why a petrol engine would emit white or light blue smoke at its exhaust pipe(**2marks**)

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b) Identify two advantages of a pressurized water cooling system over thermo-siphon cooling system (2marks)

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8 a) List two advantages of a two –stroke engine over a four –stroke petrol engine (2marks)

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b) State two types of fire extinguishers used to put out a burning carburetor (1mark)

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9a) Explain each of the following terms as used in power mechanics (3marks)

i) Detonation

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ii pre- ignition

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iii) Atomization

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10 a) State two functions of each of the following vehicle systems . (3marks)

i) suspension system

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ii) steering system

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iii) braking system

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SECTION B: (60 MARKS)

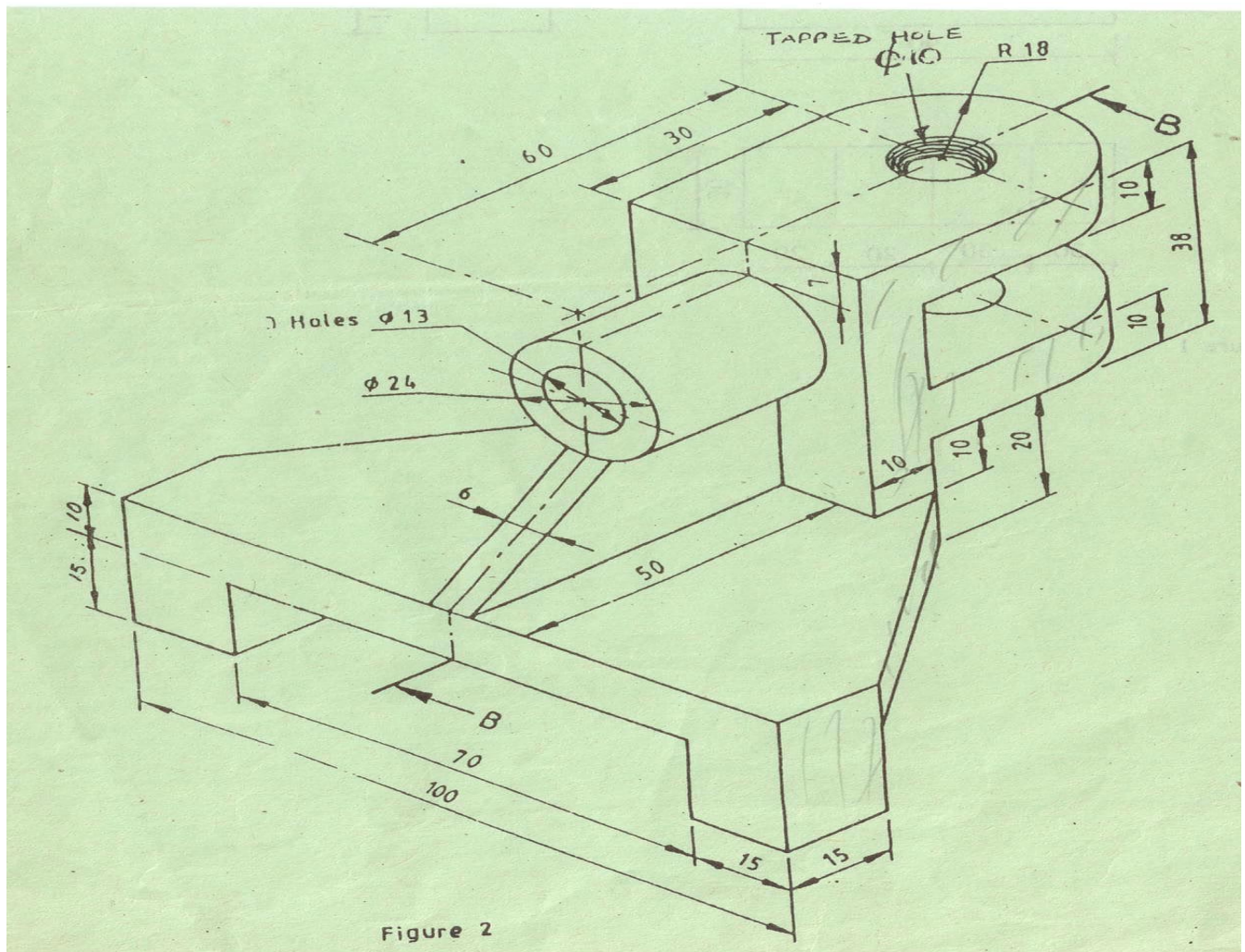
Answer question 11 and any other three questions from this section.

Candidates are advised to spend(not **more than 25 minutes**) in question 11.

11. The figure below shows an isometric drawing of a machine component. Draw FULL SIZE, in first projection the following views in the drawing paper provided. 15 marks)

- a) Sectional front elevation B-B .
- b) The plan.
- c) Include 3 important dimensions.

Do not show hidden details.



12i) State the functions of the following in a vehicle (2 marks)

- a) Final drive unit
- b) Clutch

ii) List **three** advantages of wankel rotary engine over the conventional reciprocating piston type. (3marks)

iii) Identify four differences between full-flow and By-pass filters (2marks)

iv) Using sketches explain the operation of warning light pressure gauge.

13) Draw and label four main parts of a ball bearing. (4 marks)

Draw atypical valve train diagram and label atleast six parts . (6 marks)

c) Using a well labeled diagram explain the operation of mechanical fuel pump.(5 marks)

14 a) Draw a well-based rim and label the parts .(3marks)

a) Outline the procedure to be followed when carrying out brake bleeding.(4marks)

c) State the cause of the following types of wear on a tire (3 marks)

- i) Wear at the centre of the tread
- ii) Wear on both sides of the tread
- iii) Wear on one side of the tread

d) Using sketches distinguish between king pin inclination and castor angle (3 marks)

e) List down **four** advantages of the independent front suspension over the solid I –beam axle.(2 marks)

15 a) State the function of the following in a cooling system.(3 marks)

- i) Thermostat
- ii) Pressure valve
- iii) Vacuum valve

b) List four types of manual steering boxes sketch and explain the operation of one of them. (12 marks)